1. Project Title

eBasket

---

2. Project Introduction

eBasket is a modern web-based eCommerce platform that allows businesses to sell products online while providing customers with a seamless digital shopping experience.

The system includes product browsing, secure checkout, order tracking, inventory management, and an admin dashboard for store management. It is designed to support scalable online retail operations and deliver a responsive user experience across devices.

---

3. Problem the Project Solves

Traditional retail businesses struggle with:

- Limited physical reach
- Manual inventory management
- Inefficient order tracking
- Lack of real-time analytics
- Poor customer experience

eCommerce solves these issues by providing:

- An online storefront accessible 24/7
- Automated order and inventory management
- Secure online payments
- Centralized admin control
- Customer-friendly shopping experience

---

4. Target Users

Customers

People who want to browse and purchase products online.

Store Owners

Business owners who want to sell products online and manage orders.

Admin / Platform Managers

Users responsible for managing the entire platform, products, and system settings.

---

5. Key Features

Customer Features

- User Registration & Login
- Product Browsing
- Product Categories
- Product Search & Filters
- Product Details Page

- Add to Cart
- Wishlist
- Secure Checkout
- Multiple Payment Options
- Order Tracking
- User Profile Management
- Order History
- Product Reviews & Ratings

Business Features

- Product Management
- Category Management
- Inventory Management
- Order Processing
- Sales Analytics
- Customer Management
- Coupon & Discount Management

---

6. System Modules

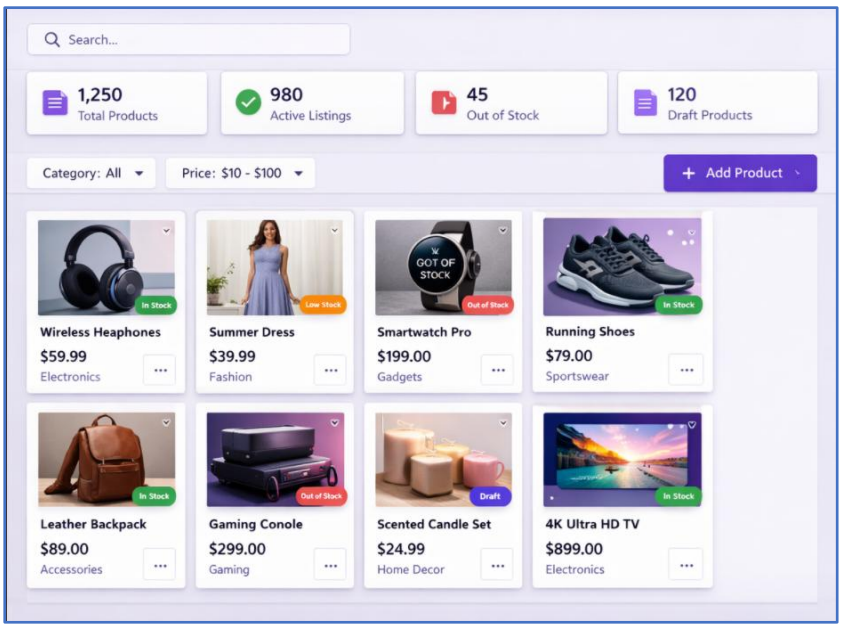
The application is divided into the following major modules:

1. Authentication Module

Handles user registration, login, logout, and password management.

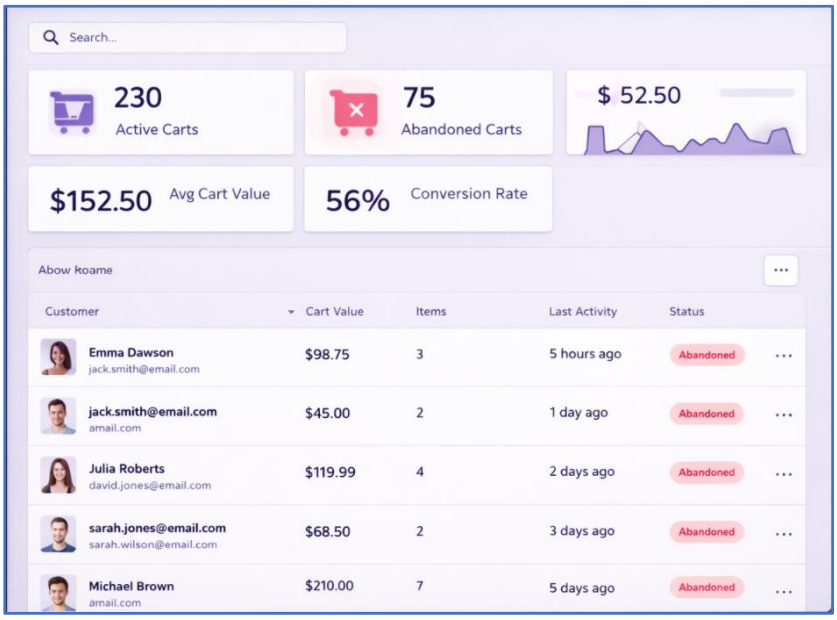
2. Product Catalog Module

Displays product listings, categories, filters, and search functionality.



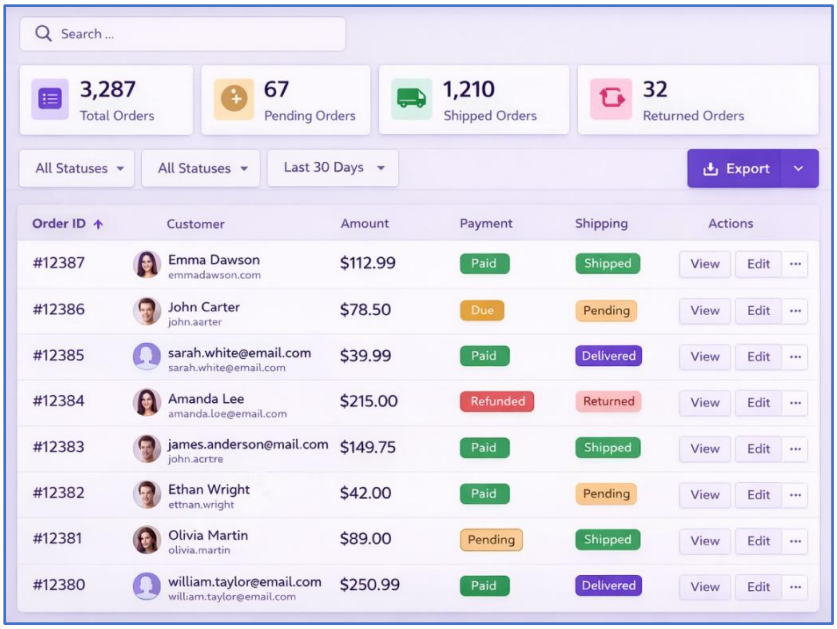
3. Shopping Cart Module

Allows customers to add, update, and remove items from their cart.



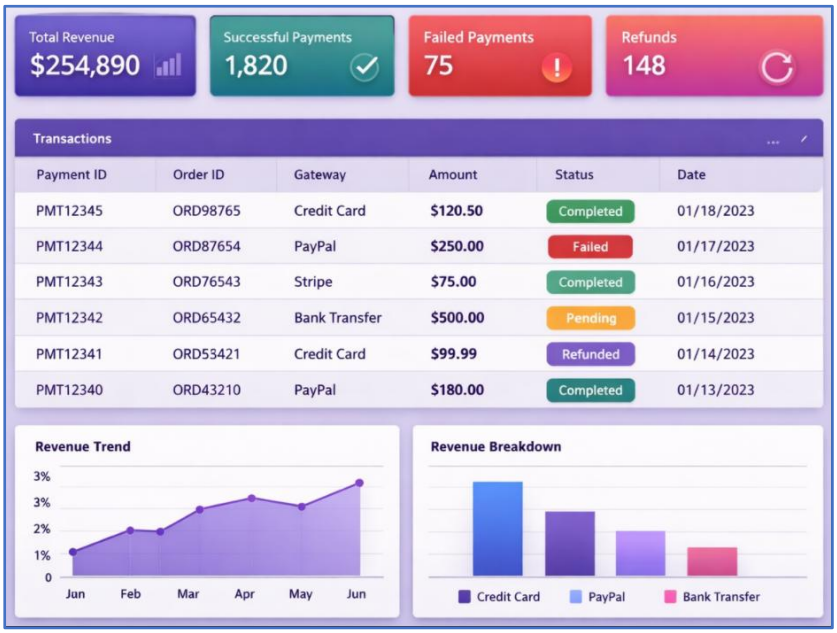
4. Order Management Module

Handles order placement, order history, order tracking, and order processing.



5. Payment Integration Module

Processes online payments using secure payment gateways.

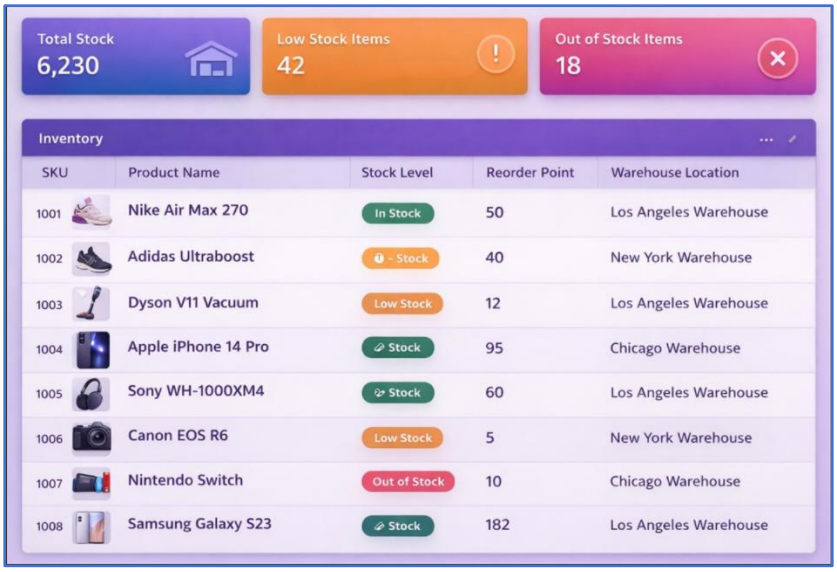


6. Admin Management Module

Allows administrators to manage products, users, orders, and reports.

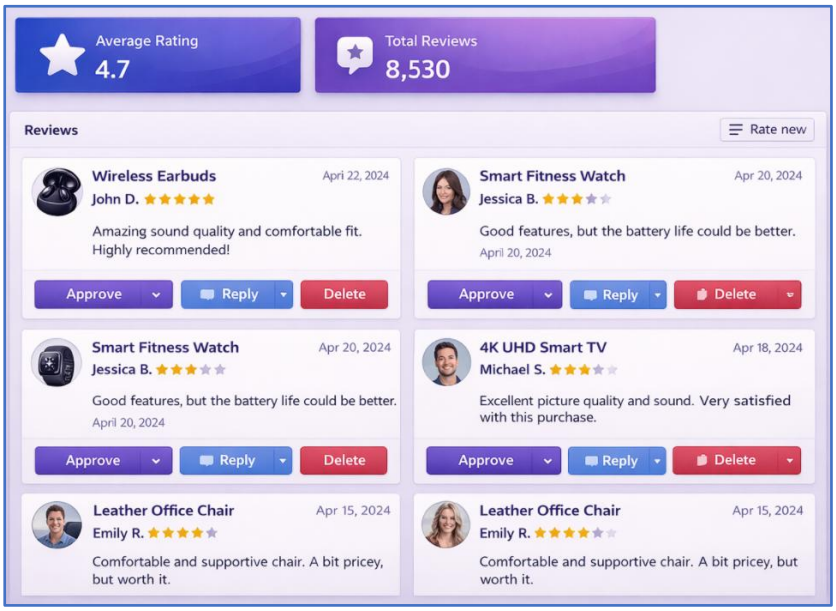
7. Inventory Management Module

Tracks product stock levels and automatically updates inventory.



8. Review & Rating Module

Allows customers to submit product reviews and ratings.



---

7. User Roles

1. Guest User

- Browse products
- View product details
- Search products

2. Registered Customer

- Add products to cart
- Place orders
- Track orders
- Manage profile
- Write product reviews

3. Admin

- Manage products
- Manage categories
- Manage users
- Manage orders
- View analytics
- Manage discounts

---

8. Core Functional Requirements

User Authentication

- Users can register using email and password
- Users can log in securely
- Password reset functionality

Product Management

- Admin can create, edit, and delete products
- Admin can manage product categories

- Products include:
    - Name
    - Description
    - Price
    - Images
    - Stock quantity

Shopping Cart

- Add products to cart
- Update product quantity
- Remove products
- View cart summary

Checkout System

- Shipping address entry
- Payment method selection
- Order confirmation

Order Processing

- Orders stored in database
- Admin updates order status:
    - Pending
    - Processing
    - Shipped
    - Delivered
    - Cancelled

Payment Integration

Supported payment options may include:

- Credit/Debit Card
- UPI
- Net Banking
- Cash on Delivery

---

9. Admin Panel Features

The Admin Dashboard allows store administrators to manage the entire platform.

Dashboard Overview

- Total sales
- Total orders
- Active users
- Revenue analytics

Product Management

- Add/Edit/Delete products
- Upload product images
- Manage pricing
- Manage stock levels

Category Management

- Create categories
- Edit categories
- Assign products to categories

Order Management

- View all orders
- Update order status
- Track shipments

Customer Management

- View registered users
- Manage customer accounts

Discount & Coupon System

- Create promotional codes
- Set expiration dates
- Apply percentage or fixed discounts

---

10. User Workflow / Journey

Customer Shopping Journey

- User visits website
- Browses product categories
- Searches or filters products
- Views product details
- Adds products to cart
- Proceeds to checkout
- Enters shipping details
- Completes payment
- Receives order confirmation
- Tracks order status

Admin Workflow

- Admin logs into dashboard
- Adds or updates products
- Manages product inventory
- Processes incoming orders
- Updates order status
- Monitors sales reports

---

11. Tech Stack

Backend

- ASP.NET Core Web API

Frontend

- Angular
- Tailwind CSS
- PrimeNG

Database

- MS SQL Server
- MongoDB
- PostgreSQL
- Redis

Architecture / Patterns

- Microservices
- MediatR
- CQRS

Communication / API Gateway

- RabbitMQ
- gRPC
- Ocelot

Cloud / DevOps / Containers

- Azure
- Docker